

Abhinav Raundhal

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EDUCATION

International Institute of Information Technology, Hyderabad

Hyderabad, India

B.Tech. in Computer Science and Engineering (Honours), CGPA: 9.64/10.0

Oct. 2022 – Jun. 2026

- **Best All-Rounder Award** (Institute Gold Medal) for excellence in academics and extracurricular activities.
- **Dean's List** in all 8 semesters (DL1: 4, DL2: 3, Merit List: 1).

PUBLICATIONS

STRinGS: Selective Text Refinement in Gaussian Splatting | *WACV 2026 (First Author)* *Project Page*

- Proposed STRinGS, a text-aware selective refinement framework for 3D Gaussian Splatting, improving text reconstruction by **63.6% OCR-CER** over prior methods.

Beyond Frontiers: Scene-Anomaly Guided Autonomous Exploration | *IROS 2026* *Project Page*

- Developed a scene anomaly-guided exploration framework that actively targets poorly reconstructed regions, improving mapping coverage and high fidelity 3D reconstruction.

EXPERIENCE

Cloudphysician | *Computer Vision Engineer* *Jun. 2026 – Present*

- Developing production-grade computer vision models and scalable ML systems for real-time critical care applications, with a focus on robust video understanding and deployment.

CVIT, IIIT Hyderabad | *Undergraduate Researcher (Computer Vision)* *May 2024 – Jun. 2026*

- Conducted research in document understanding and 3D vision, developing methods for text-aware scene reconstruction and robust handwritten document understanding.

RRC, IIIT Hyderabad | *Undergraduate Researcher (Robotics & 3D Vision)* *Jan. 2025 – Jun. 2026*

- Developed perception and exploration algorithms for autonomous navigation, emphasizing efficient 3D mapping, high-fidelity scene reconstruction, and active exploration.

ideaForge | *3D Vision Intern* *May 2025 – Jul. 2025*

- Designed a hardware-efficient 3D reconstruction pipeline for drone-based computer vision systems, optimized for resource-constrained onboard deployment.

Joly AI | *Computer Vision Intern* *Jan. 2024 – Apr. 2024*

- Developed learning-based photo enhancement models that replicate photographers' editing styles and built a React-based interface for automated image editing and workflow management.

IIIT Hyderabad | *Teaching Assistant* *Jul. 2024 – May 2026*

- Teaching Assistant for Computer Vision, Statistical Methods in AI, Computer Graphics, and Probability & Statistics; conducted tutorials, graded assignments, and mentored students.

PROJECTS

DistLearn: Federated Learning | *Distributed Systems, Privacy-Preserving ML* *Github* | *Mar. 2025 - May. 2025*

- Built a federated learning system enabling multiple clients to train locally with data privacy, aggregating model updates using methods like FedSGD, FedAvg, FedAdp and FedModCS with SSL/TLS encryption.

6DoF Pose Estimation using Differentiable Rendering | *Meshes, Graphics* *Github* | *Oct. 2024 - Dec. 2024*

- Developed a framework for 6-DoF pose estimation from meshes using differentiable rendering techniques, focusing on optimizing pose parameters by minimizing silhouette loss.

Image Inpainting and Reshuffling with NNF Algorithm | *Image Processing* *Github* | *Oct. 2024 - Dec. 2024*

- Implemented PatchMatch in Python for tasks like in-painting, watermark, and object removal, optimizing speed and memory using Nearest Neighbour Field (NNF).

SKILLS

Languages: Python, C++, C, Go, SQL | **Libraries:** PyTorch, OpenCV, NumPy, Pandas | **Tools:** Git, gRPC, Blender

LEADERSHIP & SERVICE

Cultural Secretary, IIITH Led campus cultural initiatives and coordinated institute-wide events.

Technical Secretary, Students' Parliament IIITH Revived and maintained the Students' Parliament website.

Coordinator, The Dance Crew IIITH Organized events, and participated in inter-college dance competitions.

Design Head, Electronics & Robotics Club IIITH Designed chases and tracks for robots like line follower.